

Problem 4.7

At time 0, you invest some money into an account earning 5.75% simple interest. How many years will it take to double your money?

Solution

Let the initial investment be x . Under simple interest, the accumulated amount after t years is

$$A(t) = x(1 + it),$$

where

$$i = 5.75\% = 0.0575.$$

To double the money, we need

$$2x = x(1 + 0.0575t).$$

Dividing both sides by x ,

$$2 = 1 + 0.0575t.$$

Hence,

$$\frac{2 - 1}{0.0575} = t.$$

Therefore,

$$t \approx 17.391 \text{ years.}$$